

Specializations of the Granular Prefrontal Cortex of Primates: Implications for Cognitive Processing

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ABSTRACT

The biological underpinnings of human intelligence remain enigmatic. There remains the greatest confusion and controversy regarding mechanisms that enable humans to conceptualize, plan, and prioritize, and why they are set apart from other animals in their cognitive abilities. Here we demonstrate that the basic neuronal building block of the cerebral cortex, the pyramidal cell, is characterized by marked differences in structure among primate species. Moreover, comparison of the complexity of neuron structure with the size of the cortical area/region in which the cells are located revealed that trends in the granular prefrontal cortex (gPFC) were dramatically different to those in visual cortex. More specifically, pyramidal cells in the gPFC of humans had a disproportionately high number of spines. As neuron structure determines both its biophysical properties and connectivity, differences in the complexity in dendritic structure observed here endow neurons with different computational abilities. Furthermore, cortical circuits composed of neurons with distinguishable morphologies will likely be characterized by different functional capabilities. We propose that 1. circuitry in V1, V2, and gPFC within any given species differs in its functional capabilities and 2. there are dramatic differences in the functional capabilities of gPFC circuitry in different species, which are central to the different cognitive styles of primates. In particular, the highly branched, spinous neurons in the human gPFC may be a key component of human intelligence. © 2005 Wiley-Liss, Inc.

Key words: extrastriate; cortex; pyramidal cell; human; macaque; marmoset; aotus; galago; guenon; baboon; cortical surface area

Microanatomical studies have shown that the structure of the most ubiquitous neuron in cortex, the pyramidal cell, varies markedly between different cortical areas (Fig. 1). Up to a 30-fold difference has been reported in the number of dendritic spines (the major postsynaptic sites of excitatory inputs) on neocortical pyramidal cells in the primate cerebral cortex (Elston et al., 2001, 2005h). Because pyramidal cells comprise over 70% of all neurons in cortex (DeFelipe and Fariñas, 1992), these dramatic differences in their structure are likely to influence not only cellular function, but the computational ability of the circuits they form (for reviews, see Shepherd and Greer, 1988; Churchland and Sejnowski, 1992; Koch, 1999; Mel,

1999; Segev et al., 2001; Elston, 2003a, 2006; Chklovskii et al., 2004; London and Häusser, 2005). However, the

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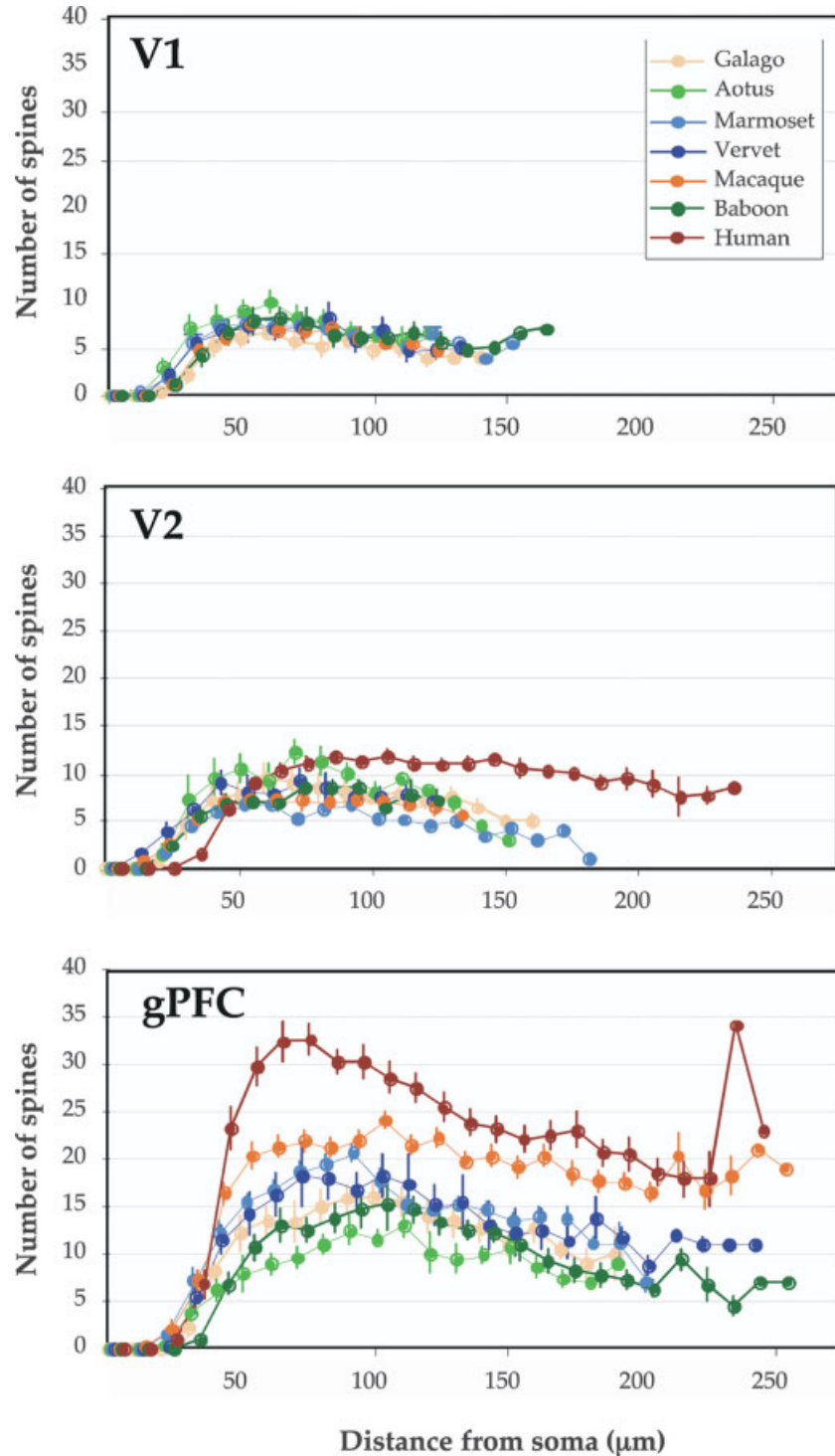


Fig. 1. Plots of spine densities along the basal dendrites of pyramidal cells in (top) the primary visual area (V1), (middle) the second visual area (V2), and (bottom) granular prefrontal cortex (gPFC) in the human, baboon, macaque monkey, vervet monkey (guenon), marmoset monkey, owl monkey (*Aotus*), and the galago. Note the differences in the profiles in the different cortical regions. In V1, the plots of spine density are remarkably tightly grouped in all primate species; in V2, there is some degree of separation, but relatively little compared with that seen in gPFC. Pyramidal cells in gPFC, particularly those in larger brained primates, are characterized by notably higher spine density along their

basal dendrites than those in V1 and V2. As each spine in cortex receives at least one asymmetric synapse (putative excitatory input), differences in spine density likely reflect variation in the number of excitatory inputs to these neurons. Data for human, macaque, marmoset, owl monkey, and galago modified from previous studies (Elston et al., 2001; Elston, 2003c; Elston et al., 2005b, 2005e, 2005h, 2005i). Data along the x-axis have been displaced to avoid cluttering of the trend lines for all species (all data were sampled at 0, 10, 20, 30, 40, 50, 60, . . . , μm from the cell body).

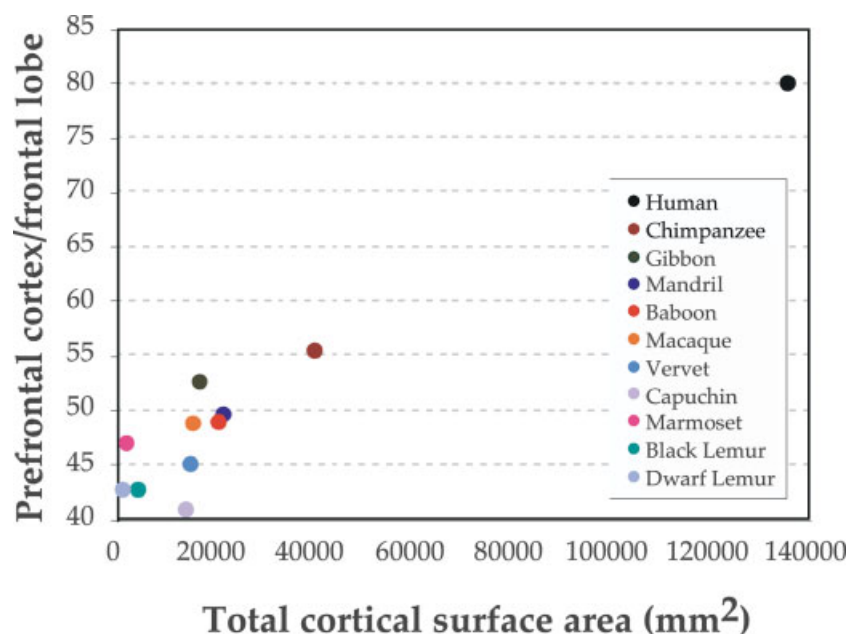


Fig. 2. Graphs illustrating the ratio of cortical surface area of granular prefrontal cortex/cortical surface area of the frontal lobe (expressed as percentage) vs. total cortical surface area (in mm²) for the human, chimpanzee, gibbon, mandrill, baboon, macaque monkey, long-tailed monkey, capuchin monkey, marmoset monkey, black lemur, and dwarf lemur. Linear regression analysis revealed a positive slope ($y =$

$0.0003x + 42.547$; $r^2 = 0.879$), suggesting that prefrontal cortex has expanded disproportionately during the evolution of these species. If gPFC had not expanded, the ratio of gPFC/frontal lobe would result in a regression with slope = 0. Prefrontal cortex is defined as granular cortex anterior to the central sulcus, as observed in Nissl preparations (Brodmann, 1913).

data reported in these studies were not related back to the size of the cortical area or brain, making it difficult to determine whether variation in their structure parallels cortical expansion.

Interspecies differences in pyramidal cell structure may parallel the relative degree of expansion of particular cortical areas, lobes, the cortical mantle, or the entire brain. Alternatively, variation in pyramidal cell structure may reflect species-specific specializations that occur irrespective of size. Establishing which of these two possibilities has occurred is essential if we are to better understand the evolution of cortical circuitry and thus specialized cortical function in different species (Kaas, 2000, 2005; Kaas and Preuss, 2003). This is particularly pertinent to the study of granular prefrontal cortex (gPFC), which has expanded considerably in humans (Fig. 2) and is thought to be important for executive cortical functions such as comprehension, planning, and perception (Goldman-Rakic, 1996; Fuster, 1997; Barbas, 2000; Rolls, 2000; Miller and Cohen, 2001).

MATERIALS AND METHODS

Human tissue (*Homo sapiens*) was obtained within 2 hr of death from a 48-year-old male who was killed instantaneously as a result of a car accident. The tissue was obtained in accordance with the guidelines for the use of human tissue of the Spanish Institutional Bioethical Committee (Spanish Council for Scientific Research). Once excised, the tissue was immersed in 4% paraformaldehyde for 24 hr (Elston et al., 2001). The other species were overdosed by lethal injection of sodium pentobarbitone and perfused intracardially with 4% paraformaldehyde.

All animals were perfused by the same investigator (G.N.E.) using the same protocol. Macaque (*Macaca fascicularis*) gPFC was taken from a 10-year-old male and occipital cortex was taken from 4.5-year-old males (Elston, 2000; Elston et al., 2005a). Marmoset (*Callithrix jacchus*) gPFC was taken from an 18-month-old male and occipital cortex was taken from 24- to 27-month-old males (Elston et al., 1999a, 2001). Tissue from the baboon (*Papio ursinus*) and vervet monkey (*Cercopithecus pygerythrus*) was obtained from wild-caught adult males of unknown age (Elston et al., 2005b, 2005c, 2005d, 2005e, 2005f, 2005g). That from the owl monkey (*Aotus trivirgatus*) was obtained from a 21-year-old animal (Elston, 2003c) and that taken from galago (*Otolemur garnetti*) was sampled from a 4-year-old animal (Elston et al., 2005h). All cases were sexually mature. When possible (human, baboon, vervet monkey, owl monkey, and galago), all tissue was sampled from a single (left) hemisphere to avoid potential hemispheric and interindividual differences in cell morphology. All experiments were performed in accordance with the relevant guidelines for the care and use of animals in each country in which the experiments were performed (United States, Australia, Spain, Japan, and South Africa).

Blocks of tissue were taken from occipital and frontal pole, including the primary (V1) and second (V2) visual areas and dorsolateral gPFC, postfixed overnight as tangential preparations, and sections (250 μ m thick) cut with the aid of a Vibratome. The sections were then prelabeled with 4,6-diamidino-2-phenylindole (Sigma D9542, St. Louis, MO) to allow visually guided injection with Lucifer Yellow (8% in 0.1 M Tris buffer, pH 7.4) by continuous negative current under fluorescence illumination. Sec-

TABLE 1 Cortical surface area (CSA), basal dendritic field area (BDFA) and total number of spines (TNS) for pyramidal cells in the primary (V1) and second (V2) visual areas and granular prefrontal cortex (gPFC)

Cortex	Variable	Species						
		<i>Otolemur</i>	<i>Callithrix</i>	<i>Aotus</i>	<i>Cercopithecus</i>	<i>Macaca</i>	<i>Papio</i>	<i>Homo</i>
V1	CSA	343 ^a	341 ^b	400 ^c	2156 ^d	1866 ^{e-i}	2559 ^d	2826 ^j
	BDFA	48362 ^k	30600 ^l	34400 ^m	44301 ⁿ	43570 ^{o,p}	65019 ^q	—
	TNS	556 ^k	699 ^l	773 ^m	795 ⁿ	735 ^{o,p}	1077 ^q	—
V2	CSA	65 ^a	98 ^r	95 ^c	—	1203 ^{e,s}	—	1454 ^j
	BDFA	64741 ^k	51700 ^{l,t}	50400 ^m	82506 ⁿ	43900 ^t	101210 ^q	86000 ^t
	TNS	1216 ^k	1240 ^{l,t}	1459 ^m	1776 ⁿ	1139 ^t	1962 ^q	2417 ^t
gPFC	CSA	—	148 ^d	—	1625 ^d	1733 ^d	2111 ^d	39287 ^d
	BDFA	118002 ^u	105022 ^t	104000 ^v	130986 ^w	133000 ^t	143672 ^w	135000 ^t
	TNS	3579 ^u	3983 ^t	—	5152 ^w	8766 ^t	5009 ^w	15138 ^t

^aRosa et al. (1997a).^bFritsches and Rosa (1996).^cTootell et al. (1985).^dBrodmann (1913).^eFelleman and Van Essen (1991).^fFlorence and Kaas (1992).^gHorton and Hocking (1996).^hLeVay et al. (1985).ⁱVan Essen et al. (1984).^jDeYoe et al. (1996).^kElston et al. (2005i).^lElston et al. (1999a).^mElston (2003b).ⁿElston et al. (2005e).^oElston and Rosa (1997).^pElston and Rosa (1998).^qElston et al. (2005b).^rRosa et al. (1997b).^sDistler et al. (1993).^tElston et al. (2001).^uElston et al. (2005h).^vElston (2003c).^wPresent results.

tions were then processed with an antibody to Lucifer Yellow [1:400,000 in stock solution, consisting of 2% bovine serum albumin (Sigma A3425), 1% Triton X-100 (BDH 30632, Poole, U.K.), 5% sucrose in 0.1 mol/l phosphate buffer], followed by a biotinylated species-specific secondary antibody (Amersham RPN 1004, Arlington Heights, IL; 1:200 in stock solution). The secondary antibodies were tagged with a streptavidin biotin-horseradish peroxidase complex (Amersham RPN1051; 1:200 in phosphate buffer) and DAB (3,3'-diaminobenzidine; Sigma D 8001) was used as the chromogen (Elston and Rosa, 1997). All cell injections were performed by two of the investigators (G.N.E. and R.B.-P.) and were standardized in all studies.

Cells were drawn in two dimensions with the aid of a camera lucida coupled with a Zeiss Axioplan microscope. Dendritic tree size was determined using NIH Image (Bethesda, MD) by drawing a convex hull joining the outermost distal tips of dendrites of each cell. Twenty horizontally projecting basal dendrites of different cells in each cortical area/species were drawn at high power (100 $\times$ oil immersion) to quantify spine densities. Spine density was determined per 10 μ m segment of dendrite as a function of distance from the cell body to the distal tips of the dendrites. All the above analysis was performed according to blind procedures. The total number of spines in the dendritic trees was calculated as the sum of the product of

average number of dendritic intersections per annul derived from Sholl analysis and the average spine density for the corresponding region of dendrites (Elston, 2001). Moderated multiple regression (Aiken and West, 1991) was used to determine significant differences between the slopes of linear regression lines. All statistical analyses were performed using SPSS software (SPSS, Chicago, IL).

RESULTS

Here we performed a systematic quantitative study of pyramidal cell structure in V1, V2, and the gPFC of human, baboon, macaque monkey, vervet monkey, owl monkey, marmoset monkey, and the galago. Specifically, we studied the size, branching structure, and total number of spines in the basal dendritic trees of cortical pyramidal cells at the base of layer III in gPFC ($n = 220$) and compared these data with those sampled from layer III pyramidal cells in V1 and V2 ($n = 232$ and 282 , respectively). These morphological parameters differed appreciably between cortical areas/regions and species (Table 1). For example, in the macaque monkey, cells in the gPFC were > 11 times more spinous than those in visual cortex. Similar comparisons in the vervet monkey revealed a six-fold difference, greater than that observed in New World marmoset monkey. Moreover, cells in the gPFC of humans were 70% more spinous than those in the next closest species (macaque monkey), three times more spinous than

those in the baboon and vervet monkey, and more than four times more spinous than those in the galago.

As each of these cortical areas/regions occupies a different absolute size in the cerebrum of the various species, it was natural to ask whether there may be some underlying trends in the regional and species differences in pyramidal cell structure related to neocortical expansion. According to the data published by Brodmann (1913), there is a 265-fold difference in the absolute size of in the gPFC of the species included in the present investigation, with the smallest observed in the marmoset (148 mm²) and the largest in the human (3,928 mm²). A 22-fold difference was observed in V2 and an 8-fold difference in V1 (Table 1). Comparison of the structure of pyramidal cells with the absolute size of the cortical area/region in which they are located revealed some interesting trends. Comparison of the size of the basal dendritic trees of pyramidal cells with the cortical surface area in V1, V2, and gPFC revealed an increase in the two variables in all cortical regions (Fig. 3A). Moreover, the slopes returned by regression analysis were remarkably similar for all three cortical regions, suggesting a common trend in primates for increasingly larger cells in increasingly larger cortical areas/regions.

However, comparison of our estimates of the total number of spines in the basal dendritic tree of the "average" neuron in each cortical area revealed different trends in gPFC, V1, and V2 (Fig. 3B). Specifically, the slope of the linear regression for the gPFC data was considerably steeper than that for either V1 or V2, suggesting that the increase in the number of spines found in the dendritic trees of pyramidal cells in gPFC during cortical expansion far exceeds that in visual cortex. To test whether this effect could be attributed to the increasing size of the dendritic trees of the neurons, we plotted the total number of spines in the dendritic tree vs. tree size in all three cortical regions (Fig. 3C). These plots revealed two important observations: there is a progressive increase in the linear regression slopes from V1 to V2 and gPFC, and there is an extraordinary amount of variance in the gPFC data not present in either V1 or V2. These data suggest that in the gPFC the number of spines in the dendritic trees of pyramidal cells more closely reflects the absolute size of this region, rather than the size of their dendritic trees.

To test whether or not these differences were significant, we performed a moderated multiple-regression analysis. By testing the relationship of any two predictors (e.g., V1/V2/gPFC and cortical surface area) on the criterion (e.g., number of spines), and by testing the product of both predictors (interaction term), we revealed significant differences in pairwise comparisons between cortical areas. A significant increase in prediction at the second test revealed a statistical difference ($P < 0.05$) between the slopes of regression lines of gPFC and V2 for comparisons between the total number of spines in the dendritic trees of pyramidal cells and the cortical surface area (Fig. 3B; $r^2_{\text{change}} = 0.112$). Significance was approached ($P = 0.057$) for the comparison between the total number of spines in the dendritic trees of pyramidal cells and the cortical surface area for gPFC and V1 (Fig. 3B; $r^2_{\text{change}} = 0.065$). Thus, human gPFC not only is considerably larger than that in other primate species (in absolute and relative terms), it is composed of pyramidal cells with highly complex dendritic trees studded with a disproportionately high number of spines (putative excitatory inputs).

DISCUSSION

In the present study, we confirm and extend previous reports of regional and species specialization in the neocortical pyramidal cell phenotype (for reviews, see Elston, 2002, 2006; Jacobs and Scheibel, 2002). We found, in many instances, dramatic differences in pyramidal cell structure among V1, V2, and the gPFC in each of the seven different primate species examined. Moreover, we found remarkable phenotypic variation in pyramidal cell structure among the different species in V1, V2, and gPFC. Comparison of the size of the pyramidal cells in V1, V2, or gPFC revealed trends for progressively larger neurons in species in which each of these cortical areas/region occupied a larger absolute cortical surface area. Furthermore, the rates of increase in cell size and the surface area of the cortical area/region were similar in V1, V2, and gPFC. Interestingly, however, we found different trends in the number of dendritic spines (putative excitatory inputs) in the dendritic trees of pyramidal cells in V1, V2, and gPFC of the different species. More specifically, pyramidal cells in species with a relatively large gPFC were disproportionately more spinous than those in species with a relatively small gPFC (cells in human gPFC were 70% more spinous than the next closest species, the macaque monkey, and three times more spinous than those in the baboon and vervet monkey). Moreover, the data reveal surprising variation in the pyramidal cell phenotype in the gPFC in primates not present in the occipital lobe. Thus, it appears as though the recent and dramatic expansion of the gPFC in primates has occurred not just by the addition of more neurons, but by the addition of more complex neurons. The evolutionary and developmental mechanisms that influence the complexity of the pyramidal cell phenotype remain to be determined, as do their genetic and epigenetic regulation (for reviews, see Nieuwenhuys, 1994; Marin-Padilla, 2001; Preuss et al., 2004; Elston, 2006).

Structure/Function Relationship

The present data reveal that the extent to which different aspects of pyramidal cell structure (size and number of spines) vary in different species depends on the cortical region studied. Of particular interest here is that the trend observed for the plots of the total number of spines in the dendritic trees vs. either cortical surface area or size of the basal dendritic trees in gPFC differs from that in V1 and V2 (Fig. 3B and C). How then might differences in the size or spine density of pyramidal cells influence their functional capabilities? While it is well known that neuron structure determines its biophysical properties (Koch, 1999), it is less well known how neuron structure may influence the functional properties of the circuits they compose. It is our contention that cortical circuits composed of pyramidal cells of different structure will be characterized by different functional capabilities, much in the same way that artificial systems composed of highly interconnected and powerful processors differ in their functional abilities to those composed of less powerful processors with fewer connections (for review, see Elston, 2003a). Several direct examples of the structure-function relationship have been demonstrated between pyramidal cell structure and cortical function. For example, differences in the size of the dendritic trees of pyramidal cells potentially influence topographic sampling strategies and mixing of inputs (Jacobs and Scheibel, 2002; Elston, 2003a). Differences in the branching complexity in the

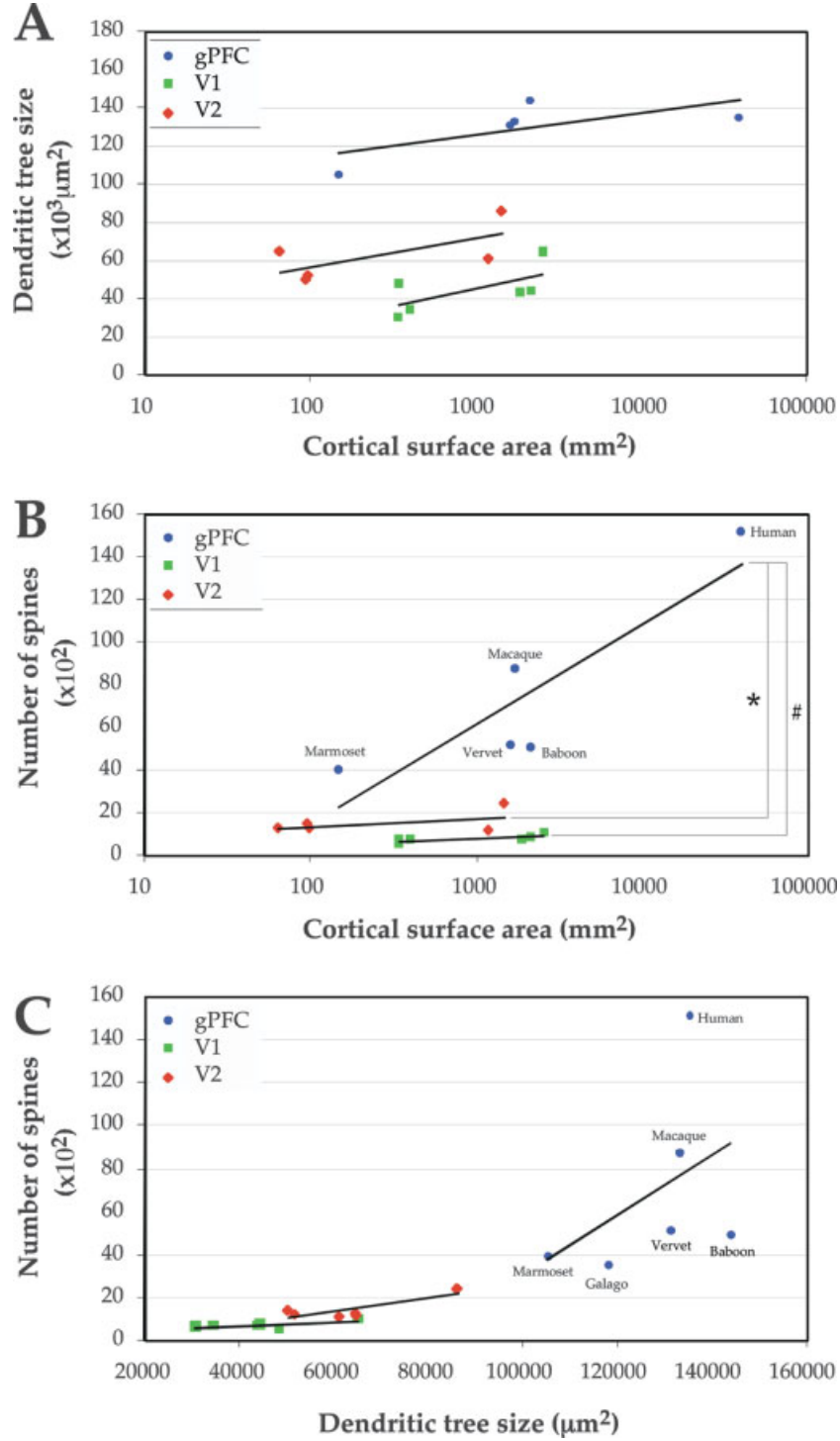


Fig. 3. Plots of the (A) size of the basal dendritic trees of pyramidal cells in the granular prefrontal cortex (gPFC), primary visual area (V1), and second visual area (V2) vs. the total cortical surface area, (B) the number of spines in the basal dendritic trees of pyramidal cells in V1, V2, and PFC vs. the total cortical surface area, and (C) the number of spines

in the basal dendritic trees of pyramidal cells in V1, V2, and gPFC vs. the size of the dendritic trees in human, baboon, macaque monkey, vervet monkey (guenon), marmoset monkey, owl monkey (*Aotus*), and the galago. Results of statistical comparisons are illustrated directly on the plots. Asterisk, $P < 0.05$; number sign, $P < 0.1$.

dendritic trees of pyramidal cells allow different degrees of compartmentalization of processing of inputs (Poirazi and Mel, 2001). Differences in the number of spines, each of which receives at least one asymmetrical synapse, in the dendritic trees of pyramidal cells reflect different numbers of excitatory inputs sampled by individual cells (Harris and Karter, 1994; Elston and DeFelipe, 2002). Thus, highly branched and spinous pyramidal cells such as those observed in human gPFC (Fig. 3B; see also Elston and Zietsch, 2006) receive more putative excitatory inputs and compartmentalize the processing of these inputs within their dendritic trees to a great extent than smaller, less spinous cells such as those in V1 and V2. These differences in neuronal structure influence their potential for plastic change (Stepanyants et al., 2002) and memory capacity (Chklovskii et al., 2004), both thought to be important for higher cortical functions.

Does this necessarily mean that more spinous cells such as those in the gPFC of humans receive inputs from a more diverse source of inputs than less spinous cells such as those in the gPFC of galagos? Reciprocity dictates that highly spinous neurons will receive more inputs than less spinous neurons, providing a basis for increased recurrent excitation (Wang, 2001; Jacobs and Scheibel, 2002; Elston, 2003a). However, very little is known of the diversity of these inputs. While studies of connectivity in the cerebral cortex are becoming increasingly more detailed (Hilgetag et al., 1996; Melchitzky et al., 2001; Collins et al., 2005; Germuska et al., 2005; Tanigawa et al., 2005), there exist relatively few standardized comparative data. To the best of our knowledge, only two such studies in gPFC have been published (Bugbee and Goldman-Rakic, 1983; Preuss and Goldman-Rakic, 1991). These authors studied patterns of connectivity in the gPFC of the New World squirrel monkey and the prosimian galago and compared them with those in the gPFC of the macaque monkey. Notable differences were detailed in the patterns of connectivity in the gPFC of the macaque and both other species. However, these studies need to be extended to include a greater diversity of species, and the patterns of connectivity quantified. A systematic comparative study of patterns of connections to the gPFC of different primate species will no doubt reveal new insights fundamental to a better understanding of cognition in primates.

There are, however, extensive data on patterns of connectivity in different regions of the cerebral cortex of the macaque monkey, which reveal regional differences in the diversity of inputs (for reviews, see Felleman and Van Essen, 1991; Young, 1993). Comparison of these data with those on regional differences in pyramidal cell structure (Lund et al., 1993; Elston et al., 1999b, 2005a; Elston, 2000; Elston and Rockland, 2002) reveals some interesting functional parallels. For example, highly branched, spinous neurons in gPFC are characterized by their tonic activity, which is sustained despite intervention from distractors; less branched and less spinous neurons in sensory association cortex are characterized by tonic activity, which desists following presentation of distractors; and the least branched and least spinous cells in V1, for example, are characterized by phasic discharge properties (Fuster and Alexander, 1971; Ashford and Fuster, 1985; Koch and Fuster, 1989; Constantinidis and Steinmetz, 1996; Miller et al., 1996; Leung et al., 2002; Sakai et al., 2002). Accepting the parallel between neuron structure and function, the present results suggest that species dif-

ferences in prefrontal functions such as conceptual thinking, prioritizing, and planning may be attributed in part at least to specializations in cortical microcircuitry. It should be relatively easy to test the predictions *in vivo*. For example, based on our data, we would expect that the gPFC of the galago, for example, would be less adept at sustaining tonic activity during presentation of distractors than that in the macaque monkey.

Methodological Considerations

Because of the difficulty in obtaining suitable material from diverse primate species for comparisons (Crick and Jones, 1993), selection of species included for study here has been somewhat fortuitous. In order to be suitable for the cell injection technique, tissue has to be obtained following surgical resection, perfusion, or postmortem. While tissue collected following surgical resection or perfusion may yield large numbers of cells (> 1,000 cells per case), tissue obtained postmortem is particularly problematic. Moreover, sampling of diverse primate species is plagued with logistical challenges; species are located in different geographical locations providing additional bureaucratic and political challenges. These experiments from which the present data are drawn were performed over a period of 10 years in five countries across four continents. Cell injection laboratories were set up in each country, and the cell injection methodology was standardized in all studies. Species selection was guided principally by the species Brodmann (1913) studied, for which there exists a consistent set of quantified data (for a translation, see Elston and Garey, 2004; see also Garey, 1994). Unfortunately, there are some glaring omissions of species in the present study, including the gorilla, chimpanzee, and the orangutan, data from which would provide crucial insights into evolutionary trends in pyramidal cell structure.

Given these limitations on the selection of cases included here for analyses, it is natural to ask whether the present results may be attributable to some form of sampling error such as interindividual variation or aging. While we cannot rule out the possibility that these sources of error may have influenced the results, it is unlikely that they could account fully for our observations. For example, in five of the seven species we injected cells in V1, V2, and gPFC in a single hemisphere, discounting the possibility of interindividual variation. Other experiments that show consistent differences in pyramidal cell structure in somatosensory, motor, and cingulate cortex in different cases make it increasingly less likely that the data presented here can be attributed to interindividual variation (Elston et al., 2005c, 2005d, 2005f, 2005g). Based on our previous cell injection studies in which we have sampled neurons from a given cortical area in up to five different individual age-matched animals, we would expect an interindividual variance in the total number of spines on pyramidal cells of no more than 30% (e.g., Elston et al., 2005a). To account for the differences reported here, interindividual variance in gPFC would have to be 22-fold greater than that in V1 and 9-fold greater than that in V2 (i.e., a variance of 521 spines in V1, 1,278 spines in V2, and 11,559 spines in gPFC).

Recently, Jacobs and colleagues revealed systematic differences in the maturation rate of pyramidal cells in different cortical areas (Travis et al., 2005), confirming and extending their initial observations in V2 and gPFC (Jacobs et al., 1997). Moreover, dendritic and spine loss is a

common occurrence in aging (Nakamura et al., 1985; Anderson and Rutledge, 1996; Duan et al., 2003). As the data presented here were sampled from animals of different age, it is logical to question whether the results can be attributed to age-related bias. While data have been published on the longevity of many primate species (Nowak, 1999), less is known of their relative developmental ages. Here we restricted our analyses to mature animals, as judged by their ages (when known), and the presence of mature secondary sexual characteristics. As the gPFC matures considerably later than visual cortex, any such error would most likely result in underestimates in the extent of differences reported between these two regions.

While estimates of the cortical surface area have been calculated for individual cortical areas V1 and V2, the estimates of the size of the gPFC include many cortical areas (Brodmann, 1909). The relative size and number of cortical areas within the gPFC may vary between species (Brodmann, 1909). While quantitative comparative data exist for the gPFC of humans and the macaque monkey (Petrides and Pandya, 2001), little or nothing has been published on the other species included here. Thus, it is possible that some of the variance in the gPFC data may be attributed to species differences in the size of the cortical areas in the gPFC. A more accurate depiction of the trends illustrated in the gPFC of different primate species is contingent on these data becoming available.

Because of the size of the dendritic trees of some pyramidal cells (up to 1 mm in diameter), it is difficult to inject large numbers of cells in transverse cortical slices and visualize their entire dendritic trees. To do so would require that cells be injected in cortical slices > 1 mm thick. Normal biological variability in the trajectory of the apical dendrite, and asymmetries in the dendritic tree, would require that sections > 1.5 mm thick be used, and cell bodies located at a depth of 0.6–0.8 mm be injected, to maximize the possibility of reconstructing the entire dendritic tree. However, it is not possible to visualize cell bodies at such a depth sufficiently well to allow injection under visual guidance, making it impossible to sample sufficient numbers of cells in each case to allow meaningful statistical comparisons. Thus, we have focused our study on the basal dendrites of pyramidal cells, as seen in tangential sections. Moreover, we have restricted the present analyses to layer III pyramidal cells. However, our observations of the basal dendrites of infragranular pyramidal cells suggest that they too differ markedly between cortical areas, and the nature of these differences reflects those observed in supragranular pyramidal cells (Elston and Rosa, 2000). Moreover, our observations of the apical dendrites of pyramidal cells, although somewhat limited, suggest that they too differ in their tangential extent, branching complexity, and spine density in much the same way as reported for the basal dendritic trees (data not shown). However, further systematic comparative studies are required to provide more detail and a better understanding of regional and species specializations in cortical circuitry.

Two other possible sources of error pertain to the determination of the cortical layer in which neurons were injected and the sampling strategy adopted for intracellular injection. In the first instance, all cortical areas included in the present study have a granular layer (layer IV). We use the nomenclature of Hassler (1966) in preference to that used by Brodmann (1909) for reasons outlined in

Casagrande and Kaas (1994) and Elston and Rosa (1997). In V1, we sampled cells immediately above the granular layer, corresponding to layer IIIC of Hassler and layer IVC α of Brodmann. In V2 and gPFC, cells were sampled at the base of layer III according to the nomenclature of both Hassler and Brodmann. The granular layer is easily distinguished in DAPI-labeled sections by virtue of the high cell density and small cell body size (see Fig. 2 of Elston and Rosa, 1997). Thus, despite differences in the overall thickness of cortex among cortical areas and species, we were able to identify the section that contained the granular layer and select the adjacent section for intracellular injection (that section closer to the cortical surface, which contained layer III).

In the second instance, DAPI-labeled cells were injected pseudorandomly. Cells are injected in a grid-like pattern allowing even sampling across cortex. Thus, pyramidal cells that project within the cortical area, to other cortical areas (ipsi- and contralateral), and to subcortical targets were all likely to have been included for analyses. Because the structure of these cell types differs even within a cortical area, some of the trends presented here may reflect variation in the proportion of these cell types among cortical areas and species. This possibility does not detract from our main findings, that is, there exist regional and species differences in pyramidal cell structure, which are likely to influence patterns of connectivity and function. Interestingly, however, while the structure of these projection-identified pyramidal cells may be distinguishable within a cortical area, the structure of each type differs markedly between cortical areas (e.g., callosally projecting or corticocortically projecting cells), leading to the conclusion that evolutionary and developmental mechanisms that determine patterns of arealization and lamination in cortex provide a stronger influence over pyramidal cell structure than the targets to which they project (Vercelli and Innocenti, 1993).

SUMMARY AND CONCLUSIONS

The granular prefrontal cortex has undergone dramatic expansion in primates. This expansion has not occurred simply by the addition of more neurons, but by the addition of more complex neurons. Specifically, there is a dramatic increase in the branching complexity and number of spines in the dendritic trees of pyramidal cells in the human gPFC. The disproportionately high number of spines in the dendritic trees of pyramidal cells in the human gPFC suggests they receive more excitatory inputs than their counterparts in other primate species. The increase in neuron number, coupled with the increasing potential to sample progressively more inputs, makes it likely that the human gPFC is characterized by more complex connectivity than that in other species. Species differences in the gPFC pyramidal cell phenotype, and the circuits they compose, are likely to influence functions associated with this region such as comprehension, planning, and perception. In particular, species differences in the gPFC pyramidal cell phenotype are likely to influence their cognitive styles.

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